Proposal

# Cooperative Multi-Anchor Beam-Steering for Uniform 3D Positioning

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**University:** Paris-Saclay University.

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## 1. Proposal

### *Problems Addressed*

Achieving robust three-dimensional (3D) indoor positioning remains a significant technical challenge in Optical Wireless Positioning (OWP). Current state-of-the-art solutions, predominantly based on Multi-Fixed LED (MFL) architectures, face inherent physical limitations when deployed in large-scale environments. The primary issue is the spatial inhomogeneity of the Signal-to-Noise Ratio (SNR); fixed Lambertian sources provide adequate coverage at the room center but suffer from severe path loss and unfavorable irradiance angles at the cell edges. Furthermore, conventional Received Signal Strength (RSS) methods exhibit a critical vulnerability known as orientation-location ambiguity, where device tilt is misinterpreted as a change in distance.

Although emerging Single Beam-Steered LED (SBL) architectures represent a promising advancement by exploiting temporal angular diversity to resolve this ambiguity, their application in large volumetric spaces presents significant challenges. A centralized single-anchor topology inherently lacks spatial redundancy, making the system highly susceptible to service outages due to blockage and suffering from radial error propagation, where angular uncertainties amplify linearly with the distance from the source.

### *Motivation*

The motivation of this research is to explore the fundamental performance limits of current 3D positioning estimation using traditional fixed configurations (MFL) and to benchmark these against Beam-Steered LED architectures. A critical objective is to identify and optimize the fusion methodologies required when scaling from a single source to a cooperative multi-anchor system, herein referred to as the MBL architecture. This investigation will follow a phased approach, commencing with comprehensive theoretical simulations and progressing to experimental validation. Notably, the comprehensive characterization of the MFL baseline, encompassing both theoretical simulation and experimental trials, presents a strategic opportunity for external collaboration with research groups possessing established fixed-LED testbeds.

### *Solution Proposed*

To overcome the aforementioned limitations, this research proposes a novel Distributed Multi-Anchor Beam-Steering (MBL) framework. Unlike centralized approaches, this solution deploys a network of independent beam-steering anchors operating in an asynchronous scanning mode. By aggregating local angular estimates through a geometric fusion strategy, the system aims to achieve uniform 3D positioning accuracy.

To rigorously benchmark the efficacy of this proposal, the study will conduct a comparative performance analysis of three distinct architectures:

**A. Architecture I: Multi-Fixed LED (MFL) - The Baseline** [Kevin: Interesting to explore the limits of 3D estimation with this architecture]

- **Configuration:** 4 and 6 fixed Lambertian LEDs located at the ceiling corners or in a symmetric grid.
- **Method:** RSS-based trilateration using **Non-Linear Least Squares (NLLS)** optimization.
- **Role:** Represents the current industry standard benchmark. It serves to quantify the degradation in positioning accuracy at the room edges and the system's sensitivity to orientation-location ambiguity when the UE is tilted.

**B. Architecture II: Single Beam-Steered LED (SBL) - The Specialist**

- **Configuration:** A single SBL unit located at the center of the ceiling.
- **Method:** A standalone Generalized Least Squares (GLS) estimator, as validated in previous work (TCOM). This method records the RSS vector corresponding to K distinct sequential beam orientations. By exploiting this temporal angular diversity, the algorithm estimates the direction and distance to the UE. This formulation treats the receiver's tilt as a common scaling factor across the K measurements, making the position estimation invariant to the UE's orientation (i.e., it inherently cancels the tilt effect) without needing to explicitly estimate the tilt angle.
- **Role:** Validates the efficacy of the GLS algorithm in resolving the orientation-location ambiguity using a single source. However, it highlights the physical limitations of a centralized topology: radial error growth (where angular uncertainties amplify with distance) and high susceptibility to service outages due to blockage (Single Point of Failure due to lack of spatial redundancy).

**C. Architecture III: Distributed Multi-Anchor Beam-Steering (MBL) - The Proposal** [Kevin: This is the extension of our SBL method, but adapted for cooperative SBL]

- **Configuration:** ≥2 independent SBL units located in a symmetric grid.
- **Operation Mode:** Asynchronous Spatial Scanning.
- **Orchestration & Fusion Strategy:**
    1. **Local Estimation:** The UE executes the tilt-invariant GLS algorithm (as defined in Architecture II) locally for each observable anchor. This yields a set of directional vectors (Lines of Position) that are robust against receiver tilt.
    2. **Distributed Fusion:** Since independent noise sources cause these N lines to be non-intersecting in 3D space, determining the optimal 3D coordinate is non-trivial. Therefore, this study establishes a comparative evaluation framework to identify the optimal fusion strategy. We will implement and benchmark a set of candidate algorithms, including but not limited to:
        - (i) Selecting the single most reliable anchor based on proximity.
        - (ii) A geometric average of all available lines.
        - (iii) A proposed method that weights each line inversely proportional to its estimated distance squared, minimizing the variance of the geometric error.

## 2. Challenge and Expected Outcomes

*Challenges*

To validate the proposed architecture, the simulation environment is designed to introduce stress conditions that typically cause conventional systems to fail. The study considers a large-scale room volume of 5m x 5m x 3m, a dimension specifically chosen to expose the path-loss limitations of fixed LEDs and the coverage limits of a single SBL.

A significant challenge involves the stochastic nature of the receiver. The simulation incorporates random UE tilt angles of up to 30° to test the system's immunity to orientation-location ambiguity. Additionally, the scenario addresses the issue of occlusion by modeling potential blockages (e.g., cylindrical human body models). The proposed MBL architecture must demonstrate the capability to maintain service continuity through spatial redundancy, seamlessly fusing data from distant visible anchors when the nearest node is obstructed.

*Expected Outcomes*

The primary outcome of this research will be a comprehensive comparative analysis demonstrating the superior performance of the Distributed MBL architecture over both MFL and Single-SBL baselines. We expect to show, through Cumulative Distribution Function (CDF) analysis, that the proposed method achieves uniform 3D positioning accuracy (< 5 cm) across 95% of the service area.

Furthermore, the study aims to identify the optimal strategy for fusing data from multiple asynchronous anchors. By evaluating a set of candidate algorithms, the research will define a robust framework for aggregating distributed angular estimates. This evaluation remains open to incorporating additional fusion techniques to ensure global minimization of the Root Mean Square Error (RMSE).

### 3. Potential Collaborations

This research offers a clear pathway for synergistic collaboration. While the theoretical framework and extensive simulations of the MBL architecture will be developed at LISV, the comparative benchmark requires experimental validation. There is a strong potential to partner with external research groups (e.g., Brunel University) that possess established Multi-Fixed LED testbeds. This collaboration would allow for a high-impact "System A vs. System B" study, contrasting the experimental baseline of traditional architectures against the advanced capabilities of the proposed distributed beam-steering system.